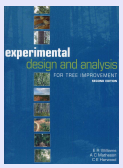
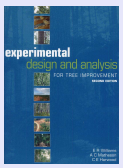


Experimental Design and Analysis for tree Improvement using R

Muhammad Yaseen

2023-04-15



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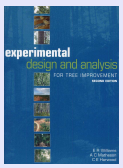


Preface

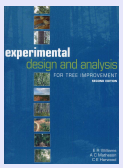
This book contains R codes and tutorials from R package eda4treeR on **Experimental Design and Analysis for tree Improvement** by E.R. Williams, C.E. Harwood and A.C. Matheson.

The development version of R package eda4treeR can be installed from github as follows:

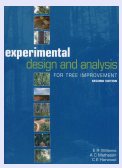
```
if (!require("remotes")) install.packages("remotes")
remotes::install_github("myaseen208/eda4treeR")
```

The stable version of R package eda4treeR can be installed from CRAN as follows:

```
install.packages("eda4treeR")
```



1Introduction



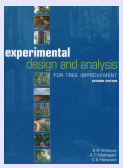
2Experimental Planning & Layout

2.1 Example 2.1 (Pg. 22)

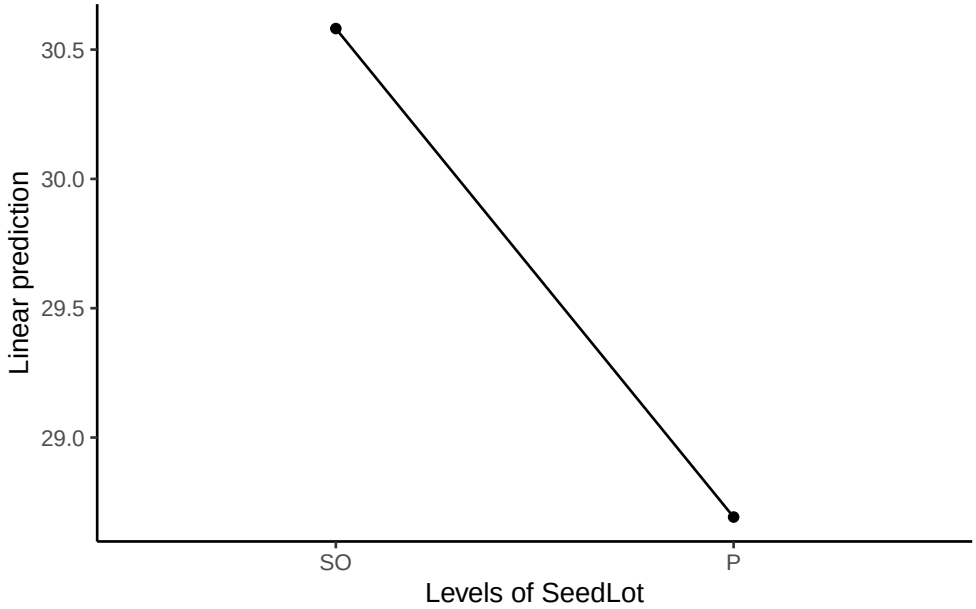
Example 2.1 (Pg. 22)

A field trial was planted to compare a seedlot derived from a seed orchard (SO) with one collected from a routine population (P). There were eight plots of each seedlot, thinned at seven year of age. Tree diameters at breast height (dbh) were measured at 15 years and plot means calculated. An R program to carry out a Completely Randomized Analysis of plot means is given below:

```
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam2.1)

# Pg. 22
fmtab2.3  <- lm(formula = dbh ~ SeedLot, data = DataExam2.1)

# Pg. 23
anova(fmtab2.3)
Analysis of Variance Table

Response: dbh
          Df Sum Sq Mean Sq F value  Pr(>F)
SeedLot    1  14.27 14.2695  3.2531 0.09284 .
Residuals 14  61.41  4.3864
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 supernova(fmtab2.3, type = 1)
 Analysis of Variance Table (Type I SS)
 Model: dbh ~ SeedLot

                            SS df      MS     F   PRE     p
 ----- --------------- | ------ -- ------ ----- ----- -----
 Model (error reduced) | 14.270  1 14.270 3.253 .1886 .0928
 Error (from model)    | 61.410 14  4.386
 ----- --------------- | ------ -- ------ ----- ----- -----
```



```
 Total (empty model)    | 75.679 15  5.045

 # Pg. 23
 emmeans(object = fmtab2.3, specs = ~ SeedLot)
 SeedLot emmean   SE df lower.CL upper.CL
 SO        30.6 0.74 14     29.0     32.2
 P         28.7 0.74 14     27.1     30.3

Confidence level used: 0.95

 emmip(object = fmtab2.3, formula = ~ SeedLot) +
   theme_classic()
```

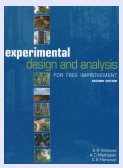


2.2 Example 2.2 (Pg. 24)

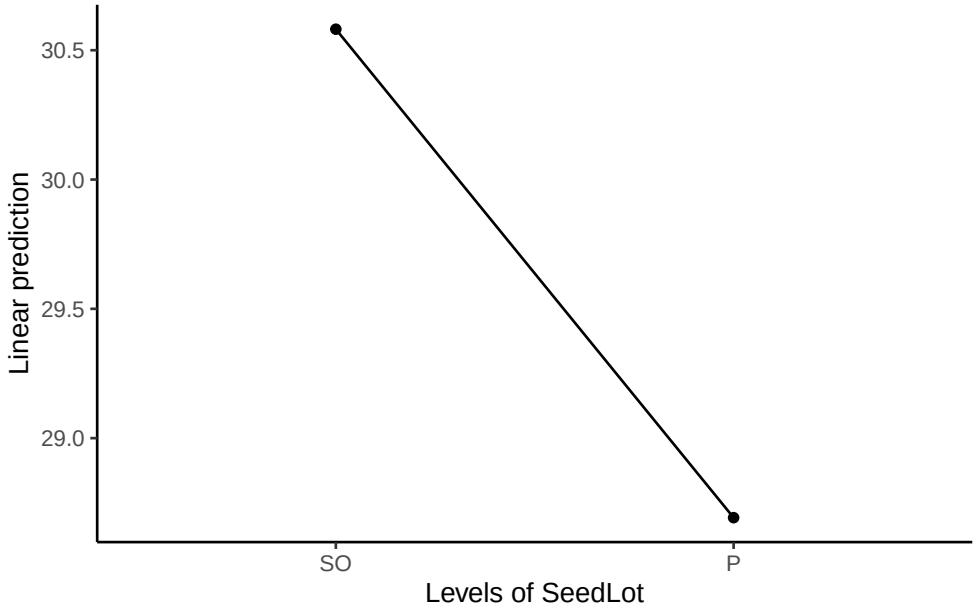
Example 2.2 (Pg. 24)

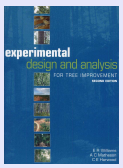
A field trial was planted to compare a seedlot derived from a seed orchard (SO) with one collected from a routine population (P). There were eight plots of each seedlot, thinned at seven year of age. Tree diameters at breast height (dbh) were measured at 15 years and plot means calculated. An `R` program to carry out a Randomized Complete Block Analysis of plot means is given below:

```
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)
```



```
library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam2.2)

# Pg. 24
fmtab2.5 <- lm(formula  = dbh ~ Blk + SeedLot, data = DataExam2.2)

# Pg. 26
anova(fmtab2.5)
Analysis of Variance Table

Response: dbh
          Df Sum Sq Mean Sq F value  Pr(>F)
Blk        7 48.867  6.9810  3.8959 0.04671 *
SeedLot    1 14.270 14.2695  7.9635 0.02570 *
Residuals  7 12.543  1.7919
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

supernova(fmtab2.5, type = 1)
Analysis of Variance Table (Type I SS)
Model: dbh ~ Blk + SeedLot

                              SS df     MS     F   PRE     p
------- --------------- | ------ -- ------ ----- ----- -----
  Model (error reduced) | 63.136  8  7.892 4.404 .8343 .0330
    Blk                 | 48.867  7  6.981 3.896 .7957 .0467
SeedLot                 | 14.270  1 14.270 7.964 .5322 .0257
  Error (from model)    | 12.543  7  1.792
------- --------------- | ------ -- ------ ----- ----- -----
  Total (empty model)   | 75.679 15  5.045

# Pg. 26
emmeans(object = fmtab2.5, specs = ~ SeedLot)
SeedLot emmean    SE df lower.CL upper.CL
SO        30.6 0.473  7     29.5     31.7
P         28.7 0.473  7     27.6     29.8

Results are averaged over the levels of: Blk
Confidence level used: 0.95

emmip(object = fmtab2.5, formula = ~ SeedLot) +
  theme_classic()
```


30.5
30.0
29.5
29.0
Linear prediction
SO
P
Levels of SeedLot



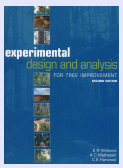
3Data Collection & Pre-Processing

3.1 Example 3.1 (Pg. 28)

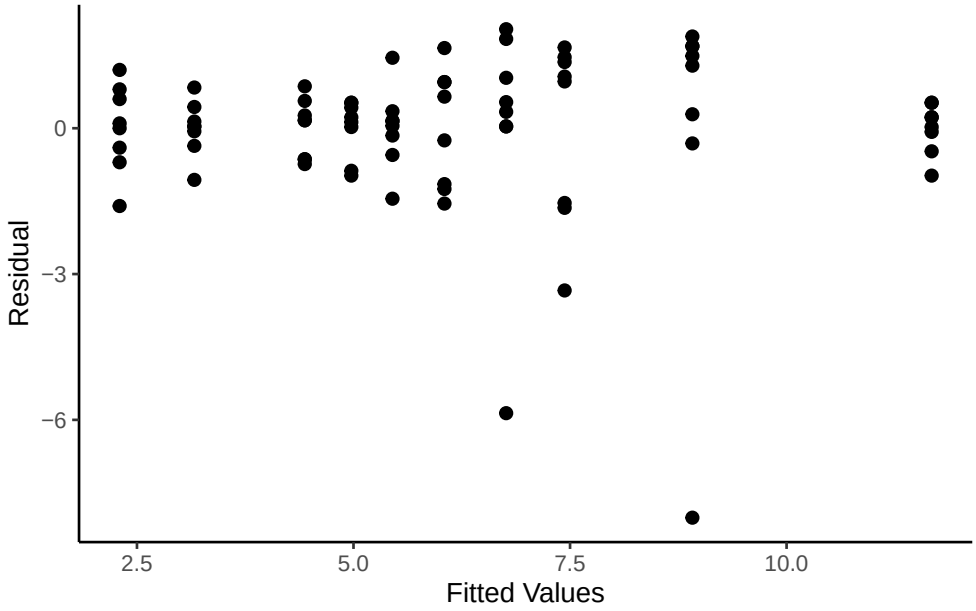
Example 3.1 (Pg. 28)

We illustrate the recommended layout for data sheets with one of the trials conducted by the Australian Centre for International Agricultural Research (ACIAR) in Queensland, Australia (Experiment 309). This was a species trial planted in 1985; survival was poor. For our example we will examine only part of the data from this experiment. Five of the species with good survival have been extracted at random. namely Acacia, Angophora, Casuarina, Melaleuca and Petalostigma. The first eight trees in each plot were selected and for convenience we have numbered the species from 1 to 5 and the trees in each plot from 1 to 8. We will assume the layout given in Figure 3.1. The data are listed in Table 3.1 and include the 4.5 year measurement of height (ht) and diameter at ground level (dgl). The plots in Table 3.1 are in field order and also replicate by replicate. It is nearly always advisable to finish collecting data from the plots of one replicate before going to the next. Usually, collecting data replicate by replicate corresponds to the natural field order, but even where an alternative exists there is an added reason for use of this collection method, namely to confound any time effect with replicates. To explain this further, suppose that an experiment is so big that the measurement team cannot collect all data in one day. The next day a different team takes over the measurement, but has a habit of reading the measuring sticks a bit short. If one replicate had been done the first day and the other replicate the second day, the height bias due to the change of measurement persons would appear in the analysis as a between-replicate effect, and although the interpretation of this effect may be difficult, the bias of the second observer would not directly affect the comparison of seedlot means.

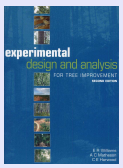
```
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam3.1)

# Pg. 28
```



```
fmtab3.3 <- lm(formula = Ht ~ Repl*SeedLot, data = DataExam3.1)

fmtab3.3ANOVA1 <-
  anova(fmtab3.3) %>%
  mutate(
  "F value" = c(anova(fmtab3.3)[1:2, 3]/anova(fmtab3.3)[3, 3]
                , anova(fmtab3.3)[4, 3]
                , NA)
                )

 # Pg. 33 (Table 3.3)
fmtab3.3ANOVA1 %>%
  mutate(
  "Pr(>F)"  = pf(q = fmtab3.3ANOVA1[ ,4]
                , df1 = fmtab3.3ANOVA1[ ,1]
                , df2 = fmtab3.3ANOVA1[4,1], lower.tail = FALSE)
                )
             Df Sum Sq Mean Sq F value           Pr(>F)
Repl          1  20.30  20.301  3.4197          0.06864 .
SeedLot       4 505.87 126.467 21.3035 0.0000000000159 ***
Repl:SeedLot  4  23.75   5.936  2.5088          0.04955 *
Residuals    70 175.61   2.509
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 # Pg. 33  (Table 3.3)
 emmeans(object  = fmtab3.3, specs = ~ SeedLot)
SeedLot      emmean    SE df lower.CL upper.CL
Acacia        10.29 0.396 70     9.50    11.08
Angophora      7.10 0.396 70     6.31     7.89
Casuarina      5.51 0.396 70     4.72     6.30
Melaleuca      4.94 0.396 70     4.15     5.73
Petalostigma   2.73 0.396 70     1.94     3.52

Results are averaged over the levels of: Repl
Confidence level used: 0.95

 # Pg. 34  (Figure 3.2)
 ggplot(mapping = aes(x = fitted.values(fmtab3.3), y = residuals(fmtab3.3)))+
 geom_point(size = 2) +
 labs(
    x = "Fitted Values"
  , y = "Residual"
   ) +
 theme_classic()
```




```
# Pg. 33 (Table 3.4)
DataExam3.1m <- DataExam3.1

DataExam3.1m[c(28, 51, 76), 5] <- NA

DataExam3.1m[c(28, 51, 76), 6] <- NA

fmtab3.4 <- lm(formula = Ht ~ Repl*SeedLot, data = DataExam3.1m)

fmtab3.4ANOVA1 <-
  anova(fmtab3.4) %>%
  mutate(
      "F value" = c(anova(fmtab3.4)[1:2, 3]/anova(fmtab3.4)[3, 3]
                    , anova(fmtab3.4)[4, 3], NA))

# Pg. 33 (Table 3.4)
fmtab3.4ANOVA1 %>%
  mutate(
        "Pr(>F)"  = pf(q = fmtab3.4ANOVA1[ ,4]
                     , df1 = fmtab3.4ANOVA1[ ,1]
                     , df2 = fmtab3.4ANOVA1[4,1], lower.tail = FALSE)
                     )
             Df Sum Sq Mean Sq F value                Pr(>F)
Repl          1  18.88  18.877 10.4201              0.001931 **
SeedLot       4 588.68 147.169 81.2367 < 0.00000000000000022 ***
Repl:SeedLot  4   7.25   1.812  0.7497              0.561661
Residuals    67  50.23   0.750
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

  # Pg. 33  (Table 3.4)
  emmeans(object  = fmtab3.4, specs = ~ SeedLot)
 SeedLot       emmean    SE df lower.CL upper.CL
 Acacia         10.87 0.224 67    10.42    11.31
 Angophora       7.76 0.231 67     7.30     8.22
 Casuarina       5.51 0.216 67     5.08     5.94
 Melaleuca       4.94 0.216 67     4.51     5.38
```



```
 Petalostigma    2.73 0.216 67      2.30      3.16

Results are averaged over the levels of: Repl
Confidence level used: 0.95
```

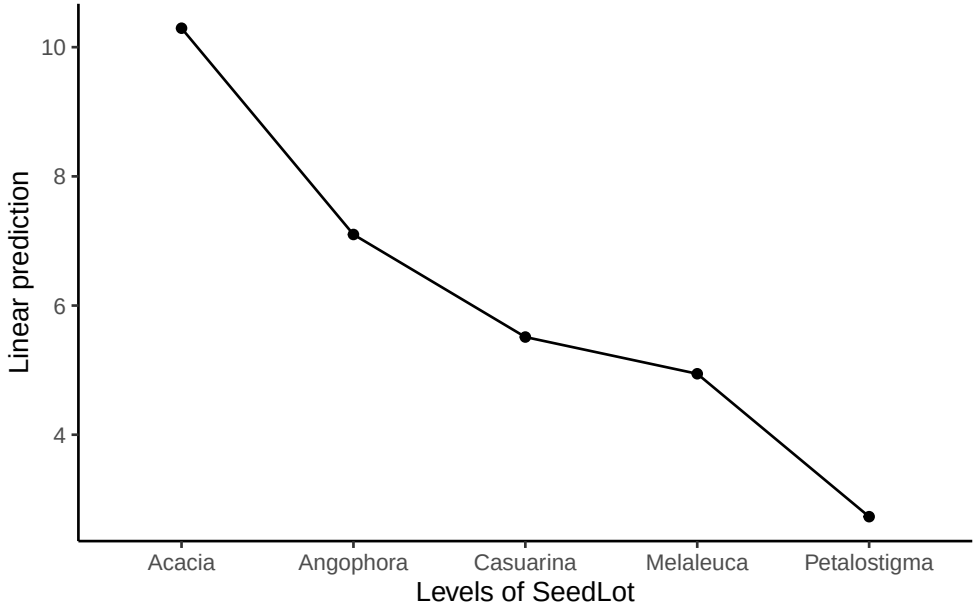
3.2 Example 3.1 (continued) (Pg. 31)

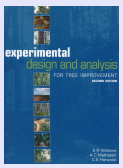
Example 3.1 (continued) (Pg. 31)

The analysis of variance table for ht is given below:

```
 library(car)

 library(dae)

 library(dplyr)

 library(emmeans)

 library(ggplot2)

 library(lmerTest)

 library(magrittr)

 library(predictmeans)

 library(supernova)

 data(DataExam3.1.1)

 # Pg. 36
 fm3.8 <- lm(formula  = Mean ~ Repl + SeedLot, data = DataExam3.1.1)

 # Pg. 40
 anova(fm3.8)
Analysis of Variance Table

Response: Mean
          Df Sum Sq Mean Sq F value   Pr(>F)
Repl       1  2.538  2.5376  3.4197 0.138108
SeedLot    4 63.234 15.8084 21.3035 0.005851 **
Residuals  4  2.968  0.7421
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 # Pg. 40
 emmeans(object = fm3.8, specs = ~ SeedLot)
 SeedLot       emmean    SE df lower.CL upper.CL
 Acacia         10.29 0.609  4     8.60    11.98
 Angophora       7.10 0.609  4     5.41     8.79
 Casuarina       5.51 0.609  4     3.82     7.20
 Melaleuca       4.94 0.609  4     3.25     6.63
 Petalostigma    2.73 0.609  4     1.04     4.42
```

```
Results are averaged over the levels of: Repl
Confidence level used: 0.95

 emmip(object = fm3.8, formula  = ~ SeedLot) +
  theme_classic()
```





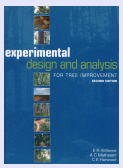
4Experimental Design

4.1 Example 4.3 (Pg. 50)

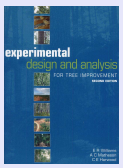
Example 4.3 (Pg. 50)

Major seed distributors such as the Australian Tree Seed Centre (ATSC) routinely conduct seed viability tests so that, when seed is dispatched, the purchaser has an indication of the germination percentage of the seed. Other areas of investigation include storage conditions, loss of viability and methods of pre-treatment for seed. so the ATSC can give seed purchasers helpful and accurate information on suitable germination procedures and viability. As part of the ATSC research program, a series of experiments was conducted in 1992 by D. Solomon on provenances of Acacia mangium to investigate methods of pre-treatment and loss of viability of stored seed. Each experiment involved six seedlots of Acacia mangium and four seed pre-treatments in a factorial design with three replicates. The seedlots were CSIRO numbers 18265, 18249, 18248, 18211, 18212 and 18217. The pre-treatments on the seed were control, nicking. iling water followed by soaking, and boiling water for one minute. For each replicate and treatment combination, 25 seeds were placed in a germination dish and the variate recorded was the number of germinants (out of 25) after a period of 21 days. Here we concentrate on the third of these experiments, conducted in August 1992.

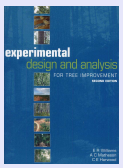
```
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam4.3)

 # Pg. 50
 fm4.2     <-
   aov(
       formula = Percent ~ Repl + Contcomp + SeedLot +
                           Treat/Contcomp + Contcomp /SeedLot +
                           Treat/ Contcomp/SeedLot
      , data    = DataExam4.3
     )
```



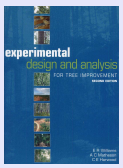
```
  # Pg. 54
  anova(fm4.2)
Analysis of Variance Table

Response: Percent
                       Df Sum Sq Mean Sq  F value                Pr(>F)
Repl                    2     35      18   0.1804             0.8355379
Contcomp                1  58542   58542 601.5217 < 0.00000000000000022 ***
SeedLot                 5   2894     579   5.9481             0.0002538 ***
Treat                   2   5300    2650  27.2295         0.00000001576 ***
Contcomp:SeedLot        5   1347     269   2.7682             0.0287571 *
Contcomp:SeedLot:Treat 10    961      96   0.9876             0.4674993
Residuals              46   4477      97
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

  # Pg. 54
  model.tables(x = fm4.2, type = "means")
Tables of means
Grand mean

51.38889

 Repl
        1     2     3
    52.33 50.67 51.17
rep 24.00 24.00 24.00

 Contcomp
    Treated Tontrol
      67.85       2
rep   54.00      18

 SeedLot
    18211 18212 18217 18248 18249 18265
       58 52.33    49 40.67 48.67 59.67
rep    12 12.00    12 12.00 12.00 12.00

 Treat
     nick  bw&s control bw1min
    40.43 49.31   51.39  64.43
rep 18.00 18.00   18.00  18.00

 Contcomp:SeedLot
         SeedLot
Contcomp  18211 18212 18217 18248 18249 18265
  Treated 77.33 69.33 63.11 53.33 64.89 79.11
  rep      9.00  9.00  9.00  9.00  9.00  9.00
  Tontrol  0.00  1.33  6.67  2.67  0.00  1.33
  rep      3.00  3.00  3.00  3.00  3.00  3.00

 Contcomp:SeedLot:Treat
, , Treat = nick

         SeedLot
Contcomp  18211 18212 18217 18248 18249 18265
  Treated 65.33 54.67 57.33 40.00 49.33 74.67
  rep      3.00  3.00  3.00  3.00  3.00  3.00
  Tontrol
```



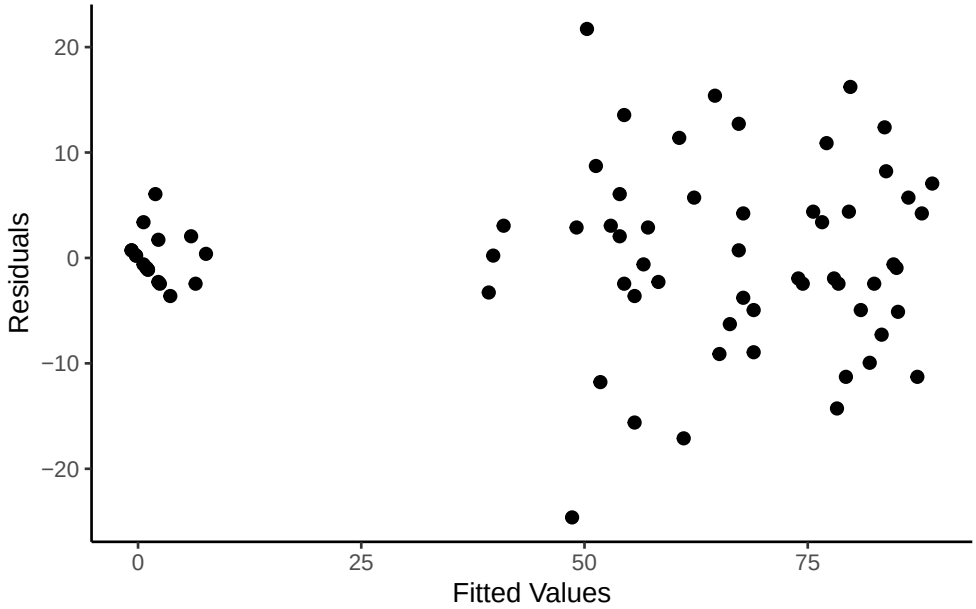
```
  rep       0.00  0.00  0.00  0.00  0.00  0.00

, , Treat = bw&s

         SeedLot
Contcomp  18211 18212 18217 18248 18249 18265
  Treated 78.67 68.00 54.67 52.00 61.33 80.00
  rep      3.00  3.00  3.00  3.00  3.00  3.00
  Tontrol
  rep      0.00  0.00  0.00  0.00  0.00  0.00

, , Treat = control

         SeedLot
Contcomp  18211 18212 18217 18248 18249 18265
  Treated
  rep      0.00  0.00  0.00  0.00  0.00  0.00
  Tontrol  0.00  1.33  6.67  2.67  0.00  1.33
  rep      3.00  3.00  3.00  3.00  3.00  3.00

, , Treat = bw1min

         SeedLot
Contcomp  18211 18212 18217 18248 18249 18265
  Treated 88.00 85.33 77.33 68.00 84.00 82.67
  rep      3.00  3.00  3.00  3.00  3.00  3.00
  Tontrol
  rep      0.00  0.00  0.00  0.00  0.00  0.00


  emmeans(object = fm4.2, specs = ~ Contcomp)
 Contcomp emmean   SE df lower.CL upper.CL
 Treated    67.9 1.34 46    65.15    70.55
 Tontrol     2.0 2.33 46    -2.68     6.68

Results are averaged over the levels of: Repl, SeedLot, Treat
Confidence level used: 0.95

  emmeans(object = fm4.2, specs = ~ SeedLot)
 SeedLot emmean   SE df lower.CL upper.CL
 18211     38.7 3.29 46     32.0     45.3
 18212     35.3 3.29 46     28.7     42.0
 18217     34.9 3.29 46     28.3     41.5
 18248     28.0 3.29 46     21.4     34.6
 18249     32.4 3.29 46     25.8     39.1
 18265     40.2 3.29 46     33.6     46.8

Results are averaged over the levels of: Repl, Treat, Contcomp
Confidence level used: 0.95

  emmeans(object = fm4.2, specs = ~ Contcomp + Treat)
 Treat   Contcomp emmean   SE df lower.CL upper.CL
 nick    Treated    56.9 2.33 46    52.21    61.57
 bw&s    Treated    65.8 2.33 46    61.10    70.46
 bw1min  Treated    80.9 2.33 46    76.21    85.57
 control Tontrol     2.0 2.33 46    -2.68     6.68

Results are averaged over the levels of: Repl, SeedLot
Confidence level used: 0.95
```



```
  emmeans(object = fm4.2, specs = ~ Contcomp + SeedLot)
 Contcomp SeedLot emmean   SE df lower.CL upper.CL
 Treated  18211    77.33 3.29 46     70.7     84.0
 Tontrol  18211     0.00 5.70 46    -11.5     11.5
 Treated  18212    69.33 3.29 46     62.7     76.0
 Tontrol  18212     1.33 5.70 46    -10.1     12.8
 Treated  18217    63.11 3.29 46     56.5     69.7
 Tontrol  18217     6.67 5.70 46     -4.8     18.1
 Treated  18248    53.33 3.29 46     46.7     60.0
 Tontrol  18248     2.67 5.70 46     -8.8     14.1
 Treated  18249    64.89 3.29 46     58.3     71.5
 Tontrol  18249     0.00 5.70 46    -11.5     11.5
 Treated  18265    79.11 3.29 46     72.5     85.7
 Tontrol  18265     1.33 5.70 46    -10.1     12.8

Results are averaged over the levels of: Repl, Treat
Confidence level used: 0.95

  emmeans(object = fm4.2, specs = ~ Contcomp + Treat + SeedLot)
 Treat   Contcomp SeedLot emmean  SE df lower.CL upper.CL
 nick    Treated  18211    65.33 5.7 46     53.9     76.8
 bw&s    Treated  18211    78.67 5.7 46     67.2     90.1
 bw1min  Treated  18211    88.00 5.7 46     76.5     99.5
 control Tontrol  18211     0.00 5.7 46    -11.5     11.5
 nick    Treated  18212    54.67 5.7 46     43.2     66.1
 bw&s    Treated  18212    68.00 5.7 46     56.5     79.5
 bw1min  Treated  18212    85.33 5.7 46     73.9     96.8
 control Tontrol  18212     1.33 5.7 46    -10.1     12.8
 nick    Treated  18217    57.33 5.7 46     45.9     68.8
 bw&s    Treated  18217    54.67 5.7 46     43.2     66.1
 bw1min  Treated  18217    77.33 5.7 46     65.9     88.8
 control Tontrol  18217     6.67 5.7 46     -4.8     18.1
 nick    Treated  18248    40.00 5.7 46     28.5     51.5
 bw&s    Treated  18248    52.00 5.7 46     40.5     63.5
 bw1min  Treated  18248    68.00 5.7 46     56.5     79.5
 control Tontrol  18248     2.67 5.7 46     -8.8     14.1
 nick    Treated  18249    49.33 5.7 46     37.9     60.8
 bw&s    Treated  18249    61.33 5.7 46     49.9     72.8
 bw1min  Treated  18249    84.00 5.7 46     72.5     95.5
 control Tontrol  18249     0.00 5.7 46    -11.5     11.5
 nick    Treated  18265    74.67 5.7 46     63.2     86.1
 bw&s    Treated  18265    80.00 5.7 46     68.5     91.5
 bw1min  Treated  18265    82.67 5.7 46     71.2     94.1
 control Tontrol  18265     1.33 5.7 46    -10.1     12.8

Results are averaged over the levels of: Repl
Confidence level used: 0.95

  DataExam4.3 %>%
    dplyr::group_by(Treat, Contcomp, SeedLot) %>%
    dplyr::summarize(Mean=mean(Percent))
# A tibble: 24 x 4
# Groups:   Treat, Contcomp [4]
   Treat Contcomp SeedLot  Mean
   <fct> <fct>    <fct>   <dbl>
 1 nick  Treated  18211    65.3
 2 nick  Treated  18212    54.7
 3 nick  Treated  18217    57.3
```



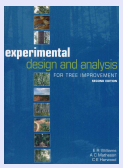
```
 4 nick   Treated   18248     40
 5 nick   Treated   18249     49.3
 6 nick   Treated   18265     74.7
 7 bw&s   Treated   18211     78.7
 8 bw&s   Treated   18212     68
 9 bw&s   Treated   18217     54.7
10 bw&s   Treated   18248     52
# i 14 more rows

    RESFIT <- data.frame(residualvalue=residuals(fm4.2),fittedvalue=fitted.values(fm4.2))

    ggplot(mapping = aes(x = fitted.values(fm4.2), y = residuals(fm4.2)))+
    geom_point(size = 2)+
    labs(
        x = "Fitted Values"
      , y = "Residuals"
      ) +
      theme_classic()
```



4.2 Example 4.3 (continued) (Pg. 57)

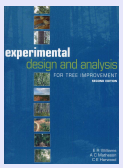
Example 4.3 (continued) (Pg. 57)

Example 4.3 was a two-factor experiment where both treatment factors were 'applied' at the same stratum level, namely the repl.row.column stratum. Sometimes it is desirable to impose treatment factors at different stratum levels, e.g. when some treatments have to be applied on a broader scale than others. We might have an RCB design for four replicates and v seedlots and want to investigate the performance of seedlots with and without a basal fertilizer application. It would be possible to apply fertilizer to two of the replicates chosen at random. Experimental structures of this type are known as split-plot designs. There is some traditional terminology associated with split-plot designs; the blocking structures are called replicates, main-plots and sub-plots, leading to the strata repl. repl.mainpl and repl.mainpl.subpl.

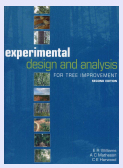


The broad-scale treatment, such as fertilizer in this example, is applied to main-plots; the seedlots would be assigned to the sub-plots. A consequence of having treatment factors at different stratum levels is that there will be separate error terms for the factors. The model for a split-plot design is of the form

```
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam4.3)

# Pg. 57
fm4.4     <-
  aov(
      formula      = Percent ~ Repl + Treat*SeedLot
    , data         = DataExam4.3 %>%
                     filter(Treat != "control")
     )

  # Pg. 57
  anova(fm4.4)
Analysis of Variance Table

Response: Percent
              Df Sum Sq Mean Sq F value      Pr(>F)
Repl           2   64.6   32.30  0.2511   0.7793606
Treat          2 5300.1 2650.07 20.6055 0.000001375 ***
SeedLot        5 4148.1  829.63  6.4507   0.0002578 ***
Treat:SeedLot 10  961.2   96.12  0.7474   0.6759614
Residuals     34 4372.7  128.61
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

  model.tables(x = fm4.4, type = "means", se = TRUE)
Tables of means
Grand mean

67.85185

 Repl
Repl
    1     2     3
69.11 66.44 68.00
```



```
 Treat
Treat
  nick    bw&s  bw1min
 56.89   65.78   80.89

 SeedLot
SeedLot
18211 18212 18217 18248 18249 18265
77.33 69.33 63.11 53.33 64.89 79.11

 Treat:SeedLot
        SeedLot
Treat    18211 18212 18217 18248 18249 18265
  nick   65.33 54.67 57.33 40.00 49.33 74.67
  bw&s   78.67 68.00 54.67 52.00 61.33 80.00
  bw1min 88.00 85.33 77.33 68.00 84.00 82.67

Standard errors for differences of means
         Repl Treat SeedLot Treat:SeedLot
        3.780 3.780   5.346         9.260
replic.    18    18       9             3

  emmeans(object = fm4.4, specs = ~ Treat)
 Treat  emmean   SE df lower.CL upper.CL
 nick     56.9 2.67 34     51.5     62.3
 bw&s     65.8 2.67 34     60.3     71.2
 bw1min   80.9 2.67 34     75.5     86.3

Results are averaged over the levels of: Repl, SeedLot
Confidence level used: 0.95

  emmeans(object = fm4.4, specs = ~ SeedLot)
 SeedLot emmean   SE df lower.CL upper.CL
 18211     77.3 3.78 34     69.7     85.0
 18212     69.3 3.78 34     61.7     77.0
 18217     63.1 3.78 34     55.4     70.8
 18248     53.3 3.78 34     45.7     61.0
 18249     64.9 3.78 34     57.2     72.6
 18265     79.1 3.78 34     71.4     86.8

Results are averaged over the levels of: Repl, Treat
Confidence level used: 0.95

  emmeans(object = fm4.4, specs = ~ Treat * SeedLot)
 Treat  SeedLot emmean   SE df lower.CL upper.CL
 nick   18211     65.3 6.55 34     52.0     78.6
 bw&s   18211     78.7 6.55 34     65.4     92.0
 bw1min 18211     88.0 6.55 34     74.7    101.3
 nick   18212     54.7 6.55 34     41.4     68.0
 bw&s   18212     68.0 6.55 34     54.7     81.3
 bw1min 18212     85.3 6.55 34     72.0     98.6
 nick   18217     57.3 6.55 34     44.0     70.6
 bw&s   18217     54.7 6.55 34     41.4     68.0
 bw1min 18217     77.3 6.55 34     64.0     90.6
 nick   18248     40.0 6.55 34     26.7     53.3
 bw&s   18248     52.0 6.55 34     38.7     65.3
 bw1min 18248     68.0 6.55 34     54.7     81.3
 nick   18249     49.3 6.55 34     36.0     62.6
```



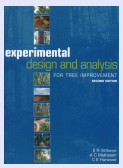
```
 bw&s    18249      61.3 6.55 34      48.0      74.6
 bw1min  18249      84.0 6.55 34      70.7      97.3
 nick    18265      74.7 6.55 34      61.4      88.0
 bw&s    18265      80.0 6.55 34      66.7      93.3
 bw1min  18265      82.7 6.55 34      69.4      96.0

Results are averaged over the levels of: Repl
Confidence level used: 0.95
```

4.3 Example 4.4 (Pg. 58)

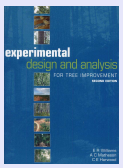
Example 4.4 (Pg. 58)

An experiment supported by the Shell Company was planted at Toolara Forest Reserve near Gympie, Queensland, in February 1987 to study the effects of irrigation and fertilizer on four seedlots of Eucalyptus grandis. Because of the difficulty in applying the irrigation and fertilizer treatments individually to each 7 x 6 plot of trees, the experiment was designed as a split-plot, with main-plot treatments, irrig and fert. There were two replicates of four main-plots each with four sub-plots. The experimental layout is given in Figure 4.3. together with sub-plot means for 34-month tree height (m).

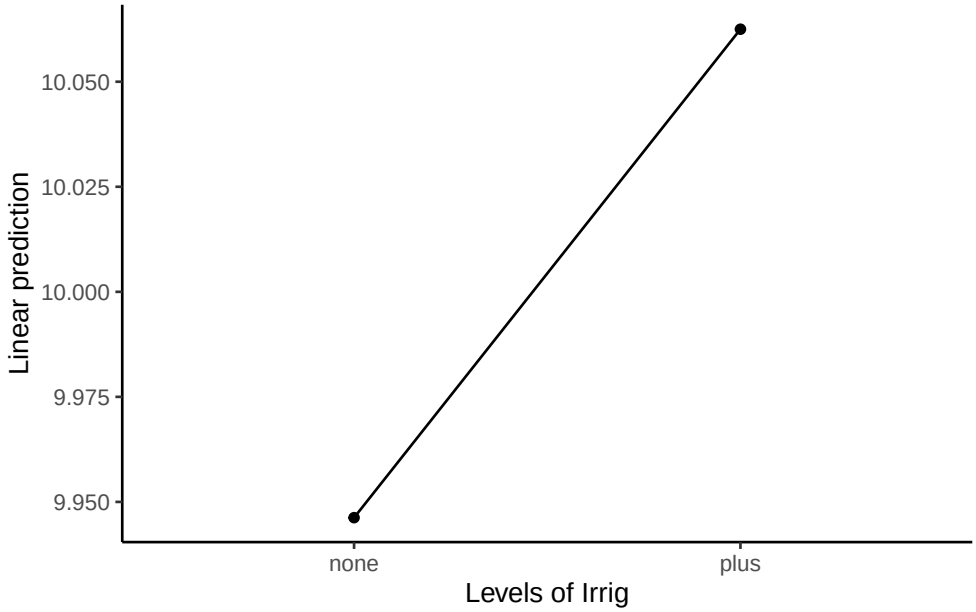
```r
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam4.4)

# Pg. 58
fm4.6     <-
  aov(
      formula      = Height ~ Rep + Irrig*Ferti*SeedDLot +
                              Error(Rep/Irrig:Ferti)
    , data         = DataExam4.4
    )

# Pg. 61
 summary(fm4.6)
```

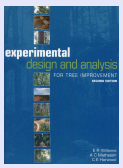


```
Error: Rep
    Df Sum Sq Mean Sq
Rep  1 0.7564  0.7564

Error: Rep:Irrig:Ferti
            Df Sum Sq Mean Sq F value   Pr(>F)
Irrig        1    0.1     0.1   0.154    0.721
Ferti        1  590.6   590.6 841.110 0.00009 ***
Irrig:Ferti  1    0.0     0.0   0.010    0.926
Residuals    3    2.1     0.7
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Error: Within
                     Df Sum Sq Mean Sq F value   Pr(>F)
SeedDLot              3  39.65  13.218  19.680 0.000063 ***
Irrig:SeedDLot        3   1.11   0.370   0.551   0.6572
Ferti:SeedDLot        3   9.95   3.317   4.938   0.0185 *
Irrig:Ferti:SeedDLot  3   1.74   0.579   0.862   0.4874
Residuals            12   8.06   0.672
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 # Pg. 61
 model.tables(x = fm4.6, type = "means")
Tables of means
Grand mean

10.00437

 Rep
Rep
     1      2
10.158  9.851

 Irrig
Irrig
  none   plus
 9.946 10.062

 Ferti
Ferti
  none   plus
 5.708 14.301

 SeedDLot
SeedDLot
                Bulahdelah Coffs Harbour seed orchard
                    10.178                     11.404
         Pomona Plantation                   Atherton
                    10.149                      8.287

 Irrig:Ferti
      Ferti
Irrig  none   plus
  none  5.635 14.257
  plus  5.781 14.344

 Irrig:SeedDLot
```



```
      SeedDLot
Irrig  Bulahdelah Coffs Harbour seed orchard Pomona Plantation Atherton
  none 10.060     11.647                     10.055             8.022
  plus 10.295     11.160                     10.242             8.552

 Ferti:SeedDLot
      SeedDLot
Ferti  Bulahdelah Coffs Harbour seed orchard Pomona Plantation Atherton
  none  5.687      6.790                      5.410             4.945
  plus 14.667     16.017                     14.887            11.630

 Irrig:Ferti:SeedDLot
, , SeedDLot = Bulahdelah

      Ferti
Irrig  none   plus
  none  5.275 14.845
  plus  6.100 14.490

, , SeedDLot = Coffs Harbour seed orchard

      Ferti
Irrig  none   plus
  none  7.125 16.170
  plus  6.455 15.865

, , SeedDLot = Pomona Plantation

      Ferti
Irrig  none   plus
  none  5.625 14.485
  plus  5.195 15.290

, , SeedDLot = Atherton

      Ferti
Irrig  none   plus
  none  4.515 11.530
  plus  5.375 11.730


 # Pg. 61
 emmeans(object = fm4.6, specs = ~ Irrig)
 Irrig emmean  SE  df asymp.LCL asymp.UCL
 none    9.95 NaN NaN       NaN       NaN
 plus   10.06 NaN NaN       NaN       NaN

Results are averaged over the levels of: Rep, Ferti, SeedDLot
Warning: EMMs are biased unless design is perfectly balanced
Confidence level used: 0.95

 emmip(object = fm4.6, formula  = ~ Irrig) +
     theme_classic()
```


10.050
10.025
10.000
9.975
9.950
Linear prediction
none
plus
Levels of Irrig



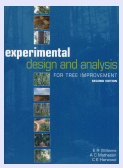
5Analysis Across Sites

5.1 Example 5.1 (Pg. 68)

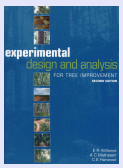
Example 5.1 (Pg. 68)

In 1985 species/provenance trials were laid out at six sites in Thailand as part of an ACIAR project extending over several years to investigate Australian multi-purpose tree species. The experimental design in each case was an RCB design with three replicates and the number of seedlots ranged from 30 to 42. Plots consisted of 5×5 trees with a $2m $\times$2 m $ spacing. Plot summary files were constructed for the 24-month. measurement according to the methods described in Chapter 3. Analyses were performed on the plot mean data at each site. Further details are given by Williams & Luangviriyasaeng (1989).

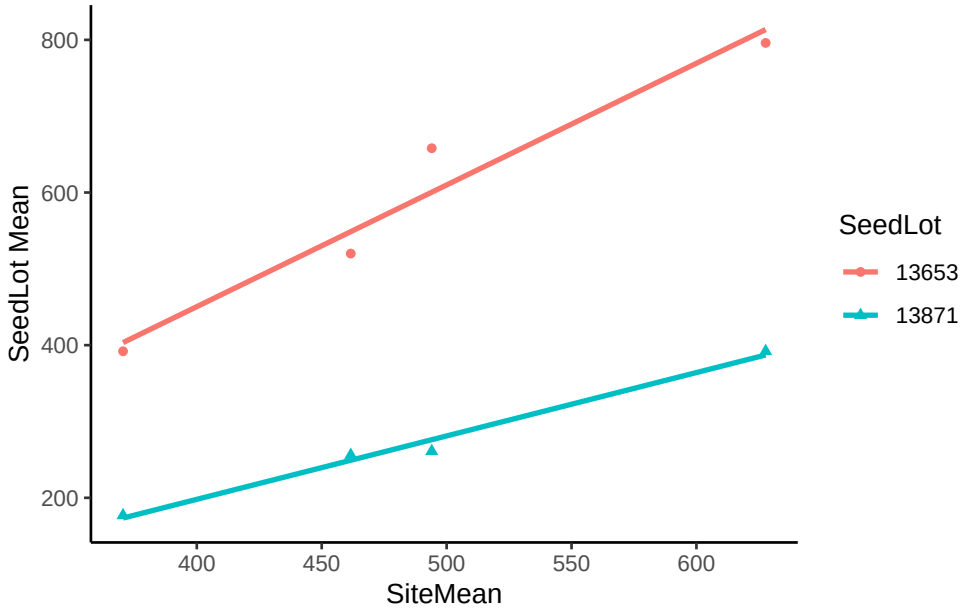
```
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam5.1)

# Pg.68
fm5.4 <- lm(formula = Ht ~ Site*SeedLot, data = DataExam5.1)

# Pg. 73
anova(fm5.4)
Analysis of Variance Table

Response: Ht
              Df  Sum Sq Mean Sq F value Pr(>F)
Site           3  919585  306528     NaN    NaN
SeedLot       26 3176289  122165     NaN    NaN
Site:SeedLot  78  707957    9076     NaN    NaN
Residuals      0       0     NaN

# Pg. 73
emmeans(object = fm5.4, specs = ~ Site)
Site       emmean  SE df lower.CL upper.CL
Rathaburi     462 NaN  0      NaN      NaN
Sai Thong     628 NaN  0      NaN      NaN
```



```
 Si Sa Ket     494 NaN  0       NaN       NaN
 Sakaerat      370 NaN  0       NaN       NaN

Results are averaged over the levels of: SeedLot
Confidence level used: 0.95

 emmeans(object = fm5.4, specs = ~ SeedLot)
 SeedLot emmean  SE df lower.CL upper.CL
 13877      365 NaN  0      NaN      NaN
 13866      353 NaN  0      NaN      NaN
 13689      559 NaN  0      NaN      NaN
 13688      546 NaN  0      NaN      NaN
 13861      627 NaN  0      NaN      NaN
 13854      628 NaN  0      NaN      NaN
 13684      660 NaN  0      NaN      NaN
 13864      422 NaN  0      NaN      NaN
 13863      586 NaN  0      NaN      NaN
 13683      770 NaN  0      NaN      NaN
 13681      695 NaN  0      NaN      NaN
 14175      438 NaN  0      NaN      NaN
 14660      521 NaN  0      NaN      NaN
 13653      592 NaN  0      NaN      NaN
 13846      440 NaN  0      NaN      NaN
 13621      384 NaN  0      NaN      NaN
 13871      272 NaN  0      NaN      NaN
 13519      422 NaN  0      NaN      NaN
 13514      369 NaN  0      NaN      NaN
 13148      273 NaN  0      NaN      NaN
 13990      282 NaN  0      NaN      NaN
 14537      780 NaN  0      NaN      NaN
 14106      772 NaN  0      NaN      NaN
 12013      616 NaN  0      NaN      NaN
 14130      422 NaN  0      NaN      NaN
 14485      123 NaN  0      NaN      NaN
 11935      273 NaN  0      NaN      NaN

Results are averaged over the levels of: Site
Confidence level used: 0.95

 ANOVAfm5.4 <- anova(fm5.4)

 ANOVAfm5.4[4, 1:3] <- c(208, 208*1040, 1040)

 ANOVAfm5.4[3, 4]   <- ANOVAfm5.4[3, 3]/ANOVAfm5.4[4, 3]

 ANOVAfm5.4[3, 5]   <- pf(
                          q = ANOVAfm5.4[3, 4]
                        , df1 = ANOVAfm5.4[3, 1]
                        , df2 = ANOVAfm5.4[4, 1]
                        , lower.tail = FALSE
                        )

 # Pg. 73
 ANOVAfm5.4
Analysis of Variance Table

Response: Ht
              Df  Sum Sq Mean Sq F value                 Pr(>F)
Site           3  919585  306528     NaN                    NaN
```



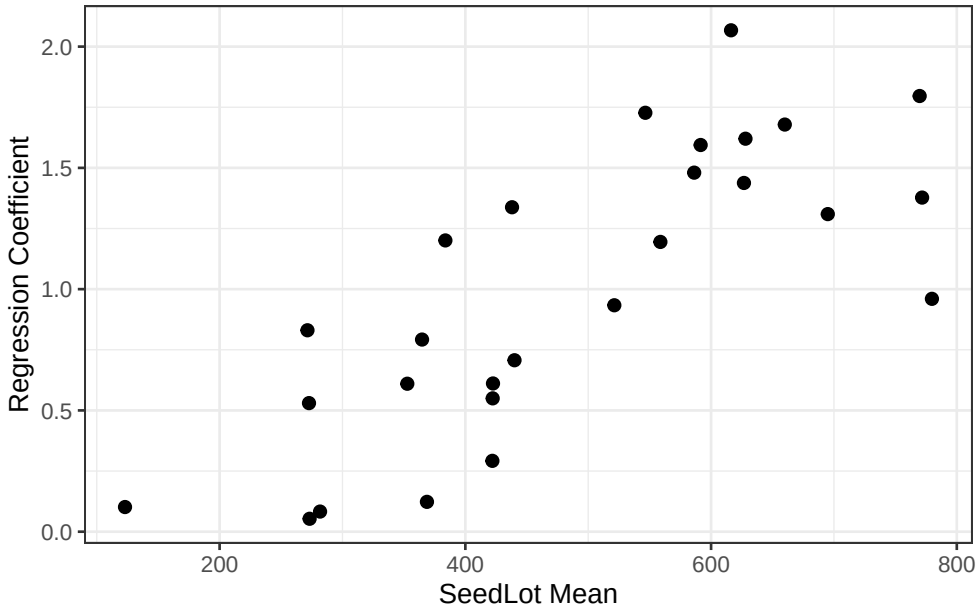
```
SeedLot        26 3176289  122165     NaN                    NaN
Site:SeedLot   78  707957    9076  8.7273 < 0.00000000000000022 ***
Residuals     208  216320    1040
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 # Pg. 80
 DataExam5.1 %>%
   filter(SeedLot %in% c("13653", "13871")) %>%
   ggplot(
     data = .
   , mapping = aes(x = SiteMean, y = Ht, color = SeedLot, shape = SeedLot)
   ) +
   geom_point() +
   geom_smooth(method = lm, se = FALSE, fullrange = TRUE)+
   theme_classic() +
   labs(
       x = "SiteMean"
     , y = "SeedLot Mean"
     )
```



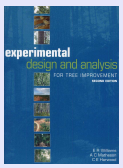

```
 Tab5.10 <-
   DataExam5.1 %>%
       summarise(Mean = mean(Ht), .by = SeedLot) %>%
     left_join(
       DataExam5.1 %>%
         nest_by(SeedLot) %>%
         mutate(fm1 = list(lm(Ht ~ SiteMean, data = data))) %>%
         summarise(Slope = coef(fm1)[2])
     , by = "SeedLot"
       )

 # Pg. 81
 Tab5.10
   SeedLot   Mean      Slope
1    11935 272.75 0.53017435
2    14485 123.00 0.10170020
3    14130 422.25 0.54976906
```

```
4     12013  616.25  2.06723798
5     14106  771.75  1.37751724
6     14537  779.75  0.96012145
7     13990  281.75  0.08298796
8     13148  273.25  0.05333546
9     13514  368.75  0.12307233
10    13519  422.00  0.29211648
11    13871  271.50  0.83048203
12    13621  383.75  1.20085607
13    13846  440.00  0.70691001
14    13653  591.50  1.59434380
15    14660  521.25  0.93353990
16    14175  438.00  1.33770745
17    13681  695.00  1.30937837
18    13683  769.75  1.79629735
19    13863  586.25  1.48034730
20    13864  422.50  0.61113857
21    13684  660.00  1.67860570
22    13854  628.00  1.62026853
23    13861  626.75  1.43784662
24    13688  546.50  1.72717652
25    13689  558.75  1.19475332
26    13866  352.75  0.61009734
27    13877  364.75  0.79221858
```

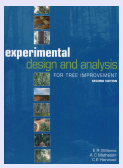
```
ggplot(data = Tab5.10, mapping = aes(x = Mean, y = Slope))+
 geom_point(size = 2) +
 theme_bw() +
 labs(
     x = "SeedLot Mean"
   , y = "Regression Coefficient"
   )
```




```
DevSS1 <-
        DataExam5.1 %>%
        nest_by(SeedLot) %>%
        mutate(fm1 = list(lm(Ht ~ SiteMean, data = data))) %>%
        summarise(SSE = anova(fm1)[2, 2]) %>%
```



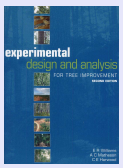
```
          ungroup() %>%
          summarise(Dev = sum(SSE)) %>%
          as.numeric()

 ANOVAfm5.4[2, 2]
[1] 3176289

 length(levels(DataExam5.1$SeedLot))
[1] 27

 ANOVAfm5.4.1 <-
   rbind(
      ANOVAfm5.4[1:3, ]
    , c(
          ANOVAfm5.4[2, 1]
        , ANOVAfm5.4[3, 2] - DevSS1
        , (ANOVAfm5.4[3, 2] - DevSS1)/ANOVAfm5.4[2, 1]
        , NA
        , NA
        )
    , c(
          ANOVAfm5.4[3, 1]-ANOVAfm5.4[2, 1]
        , DevSS1
        , DevSS1/(ANOVAfm5.4[3, 1]-ANOVAfm5.4[2, 1])
        , DevSS1/(ANOVAfm5.4[3, 1]-ANOVAfm5.4[2, 1])/ANOVAfm5.4[4, 3]
        , pf(
                q = DevSS1/(ANOVAfm5.4[3, 1]-ANOVAfm5.4[2, 1])/ANOVAfm5.4[4, 3]
              , df1 = ANOVAfm5.4[3, 1]-ANOVAfm5.4[2, 1]
              , df2 = ANOVAfm5.4[4, 1]
              , lower.tail = FALSE
              )
        )
    , ANOVAfm5.4[4, ]
   )

 rownames(ANOVAfm5.4.1) <-
   c("Site", "SeedLot", "Site:SeedLot", "  regressions", "  deviations", "Residuals")

 # Pg. 82
 ANOVAfm5.4.1
Analysis of Variance Table

Response: Ht
               Df  Sum Sq Mean Sq F value                 Pr(>F)
Site            3  919585  306528     NaN                    NaN
SeedLot        26 3176289  122165     NaN                    NaN
Site:SeedLot   78  707957    9076  8.7273 < 0.00000000000000022 ***
  regressions  26  308503   11866
  deviations   52  399454    7682  7.3863 < 0.00000000000000022 ***
Residuals     208  216320    1040
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



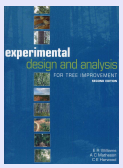
5.2 Example 5.2 (Pg. 75)

Example 5.2 (Pg. 75)

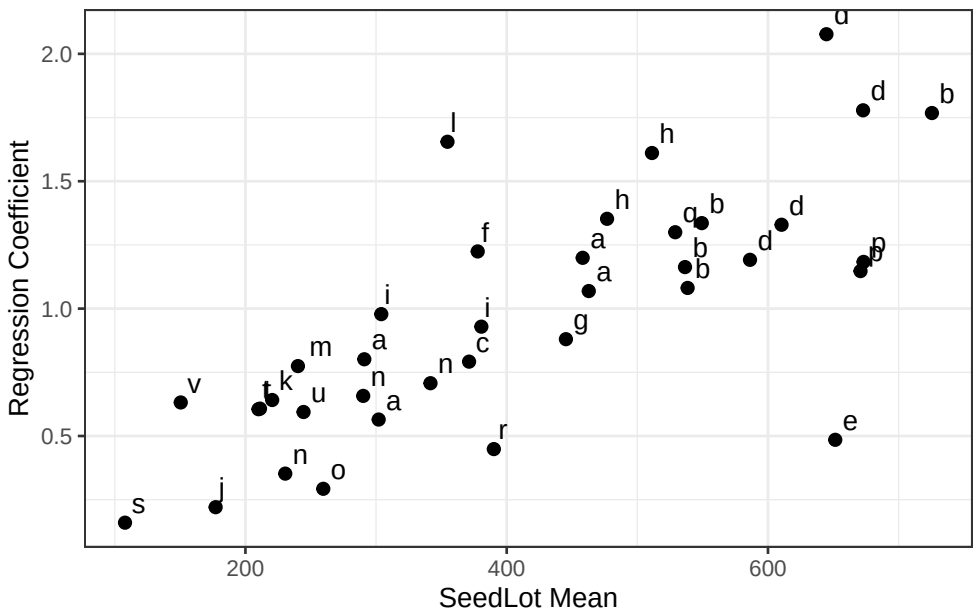
```
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam5.2)

fm5.7 <- lm(formula = Ht ~ Site*SeedLot, data = DataExam5.2)

# Pg. 77
anova(fm5.7)
Analysis of Variance Table

Response: Ht
              Df  Sum Sq Mean Sq F value Pr(>F)
Site           5 4157543  831509     NaN    NaN
SeedLot       36 4425296  122925     NaN    NaN
Site:SeedLot 150 1351054    9007     NaN    NaN
Residuals      0       0     NaN

fm5.9 <- lm(formula = Ht ~ Site*SeedLot, data = DataExam5.2)

# Pg. 77
anova(fm5.9)
Analysis of Variance Table

Response: Ht
              Df  Sum Sq Mean Sq F value Pr(>F)
Site           5 4157543  831509     NaN    NaN
SeedLot       36 4425296  122925     NaN    NaN
Site:SeedLot 150 1351054    9007     NaN    NaN
Residuals      0       0     NaN

ANOVAfm5.9 <- anova(fm5.9)

ANOVAfm5.9[4, 1:3] <- c(384, 384*964, 964)

ANOVAfm5.9[3, 4]   <- ANOVAfm5.9[3, 3]/ANOVAfm5.9[4, 3]

ANOVAfm5.9[3, 5]   <- pf(
```

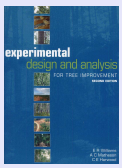


```
                              q = ANOVAfm5.9[3, 4]
                            , df1 = ANOVAfm5.9[3, 1]
                            , df2 = ANOVAfm5.9[4, 1]
                            , lower.tail = FALSE
                            )

 # Pg. 77
 ANOVAfm5.9
Analysis of Variance Table

Response: Ht
              Df  Sum Sq Mean Sq F value                Pr(>F)
Site           5 4157543  831509     NaN                   NaN
SeedLot       36 4425296  122925     NaN                   NaN
Site:SeedLot 150 1351054    9007  9.3434 < 0.00000000000000022 ***
Residuals    384  370176     964
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 Tab5.14 <-
   DataExam5.2 %>%
       summarise(Mean = mean(Ht, na.rm = TRUE), .by = SeedLot) %>%
     left_join(
       DataExam5.2 %>%
         nest_by(SeedLot) %>%
         mutate(fm2 = list(lm(Ht ~ SiteMean, data = data))) %>%
         summarise(Slope = coef(fm2)[2])
     , by = "SeedLot"
       )

 # Pg. 81
 Tab5.14
# A tibble: 37 x 3
   SeedLot  Mean Slope
   <fct>   <dbl> <dbl>
 1 13877    291  0.801
 2 13866    302  0.565
 3 13689    463. 1.07
 4 13688    458. 1.20
 5 13861    538. 1.08
 6 13854    536. 1.16
 7 13686    726. 1.77
 8 13684    549. 1.34
 9 13864    371. 0.792
10 13863    586. 1.19
# i 27 more rows

 DevSS2 <-
         DataExam5.2 %>%
         nest_by(SeedLot) %>%
         mutate(fm2 = list(lm(Ht ~ SiteMean, data = data))) %>%
         summarise(SSE = anova(fm2)[2, 2]) %>%
         ungroup() %>%
         summarise(Dev = sum(SSE)) %>%
         as.numeric()

 ANOVAfm5.9.1 <-
   rbind(
      ANOVAfm5.9[1:3, ]
```



```
    , c(
          ANOVAfm5.9[2, 1]
        , ANOVAfm5.9[3, 2] - DevSS2
        , (ANOVAfm5.9[3, 2] - DevSS2)/ANOVAfm5.9[2, 1]
        , NA
        , NA
        )
    , c(
          ANOVAfm5.9[3, 1]-ANOVAfm5.9[2, 1]
        , DevSS2
        , DevSS2/(ANOVAfm5.9[3, 1]-ANOVAfm5.9[2, 1])
        , DevSS2/(ANOVAfm5.9[3, 1]-ANOVAfm5.9[2, 1])/ANOVAfm5.9[4, 3]
        , pf(
              q = DevSS2/(ANOVAfm5.9[3, 1]-ANOVAfm5.9[2, 1])/ANOVAfm5.9[4, 3]
            , df1 = ANOVAfm5.9[3, 1]-ANOVAfm5.9[2, 1]
            , df2 = ANOVAfm5.9[4, 1]
            , lower.tail = FALSE
            )
        )
    , ANOVAfm5.9[4, ]
   )

 rownames(ANOVAfm5.9.1) <-
   c("Site", "SeedLot", "Site:SeedLot", "  regressions", "  deviations", "Residuals")

 # Pg. 82
 ANOVAfm5.9.1
Analysis of Variance Table

Response: Ht
               Df  Sum Sq Mean Sq F value                Pr(>F)
Site            5 4157543  831509     NaN                   NaN
SeedLot        36 4425296  122925     NaN                   NaN
Site:SeedLot  150 1351054    9007  9.3434 < 0.00000000000000022 ***
  regressions  36  703203   19533
  deviations  114  647851    5683  5.8951 < 0.00000000000000022 ***
Residuals     384  370176     964
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 Code <- c("a","a","a","a","b","b","b","b","c","d","d","d","d","e","f","g",
      "h","h","i","i","j","k","l","m","n","n","n","o","p","p","q","r",
      "s","t","t","u","v")

 Tab5.14$Code <- Code

 ggplot(data = Tab5.14, mapping = aes(x = Mean, y = Slope))+
  geom_point(size = 2) +
  geom_text(aes(label = Code), hjust = -0.5, vjust = -0.5)+
  theme_bw() +
  labs(
      x = "SeedLot Mean"
    , y = "Regression Coefficient"
    )
```


Regression Coefficient
2.0
1.5
1.0
0.5
200
400
600
SeedLot Mean



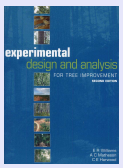
6 Variance Components & Genetics Concepts

6.1 Example 6.2 (Pg. 94)

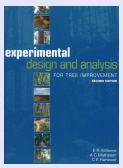
Example 6.2 (Pg. 94)

A progeny trial of Acacia mangium was planted at Segaluid, Sabah, by the Sabah. Forest Research Centre in 1994. The trial was designed to test 48 open-pollinated families collected from natural provenances in Papua New Guinea (PNG, 41 families) and far north Queensland (five families) and two families of the land. race that had developed in Sabah after introduction of A. mangium in the 1960s. Based on the results of many previous trials (Harwood & Williams 1992), it was expected that the Sabah and Queensland families would perform more poorly than those from PNG. The trial was set out as an RCB design with four replicates each containing 48 five-tree plots. Spacing was $3m \times 3m$ between trees, and an external perimeter row surrounded the trial. Diameter at breast height (dbh) and height (ht) measurements were taken in 1997, 36 months after planting.

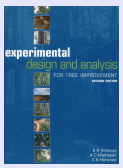
```
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam6.2)

DataExam6.2.1 <-
    DataExam6.2 %>%
    filter(Province == "PNG")

# Pg. 94
fm6.3 <-
     lm(
          formula = Dbh.mean ~ Replication + Family
        , data    = DataExam6.2.1
       )

b    <- anova(fm6.3)
```



```
HM       <- function(x){length(x)/sum(1/x)}

w        <- HM(DataExam6.2.1$Dbh.count)

S2       <- b[["Mean Sq"]][length(b[["Mean Sq"]])]

Sigma2t <- mean(DataExam6.2.1$Dbh.variance)

sigma2m <- S2-(Sigma2t/w)

fm6.3.1 <-
  lmer(
      formula    = Dbh.mean ~ 1 + Replication + (1|Family)
    , data       = DataExam6.2.1
    , REML       = TRUE
    )

# Pg. 104
# summary(fm6.3.1)
varcomp(fm6.3.1)
                     vcov     SE  2.5 % 97.5 %
Family.(Intercept) 0.2584 0.1286 0.0538 0.5767
residual           1.1667 0.1506 0.8954 1.4774

sigma2f <- 0.2584

h2 <- (sigma2f/(0.3))/(Sigma2t + sigma2m + sigma2f)

cbind(hmean = w, Sigma2t, sigma2m, sigma2f, h2)
        hmean  Sigma2t   sigma2m sigma2f        h2
[1,] 4.408602 3.920732 0.2773606  0.2584 0.1932761

fm6.4 <-
  lm(
      formula     = Dbh.mean ~ Replication+Family
    , data         = DataExam6.2
    )

b     <- anova(fm6.4)

HM       <- function(x){length(x)/sum(1/x)}

w        <- HM(DataExam6.2$Dbh.count)

S2       <- b[["Mean Sq"]][length(b[["Mean Sq"]])]

Sigma2t <- mean(DataExam6.2$Dbh.variance)

sigma2m <- S2-(Sigma2t/w)

fm6.4.1 <-
lmer(
      formula    = Dbh.mean ~ 1 + Replication + Province + (1|Family)
    , data       = DataExam6.2
    , REML       = TRUE
    )

# Pg. 107
varcomp(fm6.4.1)
```

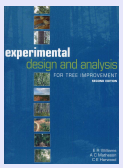


```
                      vcov     SE  2.5 % 97.5 %
Family.(Intercept) 0.3514 0.1358 0.1203 0.6361
residual           1.0951 0.1304 0.8584 1.3634

 sigma2f <- 0.3514

 h2 <- (sigma2f/(0.3))/(Sigma2t+sigma2m+sigma2f)

 cbind(hmean = w, Sigma2t, sigma2m, sigma2f, h2)
        hmean  Sigma2t  sigma2m sigma2f        h2
[1,] 4.451314 3.860156 0.227873  0.3514 0.2638477

 fm6.7.1 <-
   lmer(
       formula    = Dbh.mean ~ 1+Replication+(1|Family)
     , data       = DataExam6.2.1
     , REML       = TRUE
     )

 # Pg. 116
 varcomp(fm6.7.1)
                      vcov     SE  2.5 % 97.5 %
Family.(Intercept) 0.2584 0.1286 0.0538 0.5767
residual           1.1667 0.1506 0.8954 1.4774

 sigma2f[1] <- 0.2584

 fm6.7.2<-
  lmer(
       formula    = Ht.mean ~ 1 + Replication + (1|Family)
     , data       = DataExam6.2.1
     , REML       = TRUE
     )

 # Pg. 116
 varcomp(fm6.7.2)
                      vcov     SE  2.5 % 97.5 %
Family.(Intercept) 0.2711 0.1243 0.0743 0.5794
residual           1.0549 0.1362 0.8097 1.3359

 sigma2f[2] <- 0.2711

 fm6.7.3 <-
   lmer(
       formula    = Sum.means ~ 1 + Replication + (1|Family)
     , data       = DataExam6.2.1
     , REML       = TRUE
     , control    = lmerControl()
     )

 # Pg. 116
 varcomp(fm6.7.3)
                      vcov     SE  2.5 % 97.5 %
Family.(Intercept) 0.8729 0.3907 0.2553 1.8421
residual           3.2428 0.4186 2.4888 4.1063

 sigma2f[3] <- 0.873

 sigma2xy   <- 0.5*(sigma2f[3]-sigma2f[1]-sigma2f[2])
```



```
GenCorr <- sigma2xy/sqrt(sigma2f[1]*sigma2f[2])

cbind(S2x = sigma2f[1], S2y = sigma2f[2], S2.x.plus.y = sigma2f[3], GenCorr)
        S2x    S2y S2.x.plus.y   GenCorr
[1,] 0.2584 0.2711       0.873 0.6489119
```

7Incomplete Block Designs



8Analysis of Generalized Lattice Designs

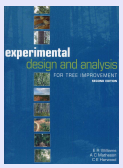
8.1 Example 8.1 (Pg. 141)

Example 8.1 (Pg. 141)

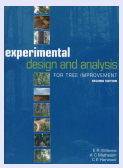
In the early 1990s Khongsak Pinyopusarerk of CSIRO Forestry and Forest Products initiated a far-reaching study of Casuarina equisetifolia. This is a nitrogen-fixing tree of considerable social, economic and environmental importance in tropical/ subtropical littoral zones of Asia, the Pacific and Africa. Provenance collections were obtained from 18 countries and, with this material, more than 40 trials were laid out in 20 countries. The number of seedlots included in each trial varied, depending on the suitability and size of the planting sites for the available material. One of the trials, in Weipa, northern Queensland, contained all the available seedlots and is the example used here. Further details are given by Pinyopusarerk & Williams (2000).

```r
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam8.1)

# Pg. 141
fm8.4 <-
  aov(
    formula = dbh ~ inoc + Error(repl/inoc) + inoc*country*prov
  , data    = DataExam8.1
     )

# Pg. 150
summary(fm8.4)
```

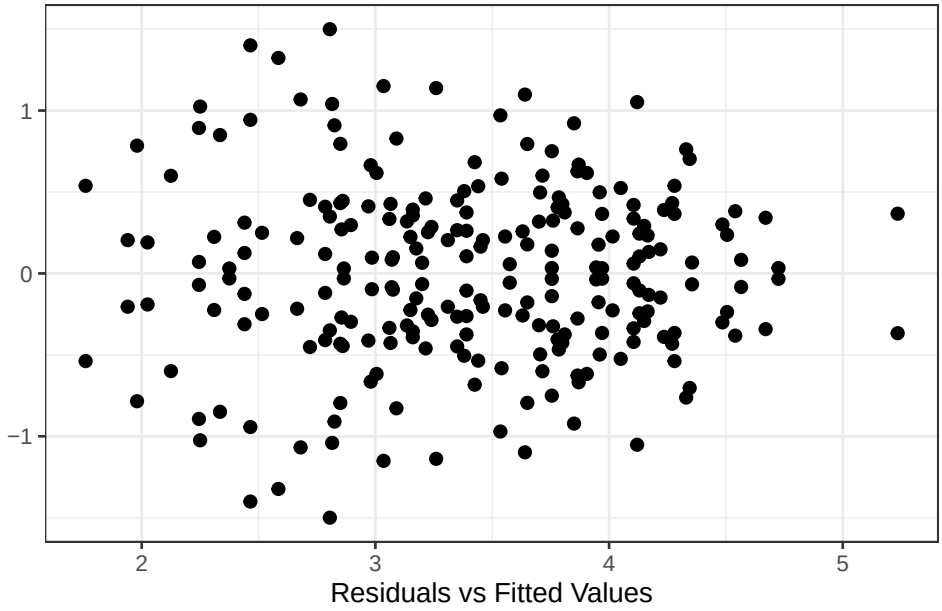
```
Error: repl
          Df Sum Sq Mean Sq F value Pr(>F)
inoc       1 11.542  11.542   11.46 0.0773 .
Residuals  2  2.014   1.007
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



```
Error: Within
               Df Sum Sq Mean Sq F value        Pr(>F)
country        17  54.62   3.213   5.305 0.0000000159 ***
prov           41  18.61   0.454   0.749         0.854
inoc:country   17  10.07   0.592   0.978         0.487
inoc:prov      41  21.46   0.523   0.864         0.698
Residuals     116  70.26   0.606
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 # Pg. 150
 model.tables(x = fm8.4, type = "means")
Tables of means
Grand mean

3.40411

 inoc
    7 weeks  1 week
      3.625   3.183
rep 118.000 118.000

 country
     India Vietnam  Egypt  Kenya   Fiji Thailand Malaysia Philippines Australia
     3.575   3.276  2.498  3.491  2.612    3.841    4.031       3.612     2.631
rep 24.000  20.000 12.000 32.000 12.000   16.000   36.000      12.000    16.000
     PNG Solomon Is. Mauritius Sri Lanka  Guam  China Puerto Rico Vanuatu Benin
    3.65       3.699     3.122     3.243 2.342  3.686       3.345   2.762 3.342
rep 4.00       8.000     4.000    12.000 4.000 12.000       4.000   4.000 4.000

 prov
        1     2    3    4     5     6     7     8    10    11   12   13    14
    2.623 4.013 3.71 3.27 3.404 3.093 3.701 3.541 3.371 3.301 3.18 3.37 3.404
rep 4.000 4.000 4.00 4.00 4.000 4.000 4.000 4.000 4.000 4.000 4.00 4.00 4.000
       15   16    17    18   19    20    21   22   23    24    25    26   27
    3.595 3.43 3.275 3.085 3.66 3.382 3.235 3.46 3.08 3.555 3.918 3.648 3.43
rep 4.000 4.00 4.000 4.000 4.00 4.000 4.000 4.00 4.00 4.000 4.000 4.000 4.00
       28    29    30    31    32    33    34    35    36    37    38   39   40
    2.905 3.708 3.196 3.761 3.416 3.178 2.958 3.636 3.376 3.404 3.252 3.15 3.81
rep 4.000 4.000 4.000 4.000 4.000 4.000 4.000 4.000 4.000 4.000 4.000 4.00 4.00
       41    42    45   46    47    48    50    51    52    53    54   55    56
    3.195 3.613 3.518 2.76 3.733 3.605 3.404 3.685 3.235 3.755 3.605 2.74 3.662
rep 4.000 4.000 4.000 4.00 4.000 4.000 4.000 4.000 4.000 4.000 4.000 4.00 4.000
       57    58    59    60    61    62    63
    3.408 3.404 3.404 3.528 3.178 3.506 3.418
rep 4.000 4.000 4.000 4.000 4.000 4.000 4.000

 inoc:country
         country
inoc      India  Vietnam Egypt  Kenya  Fiji   Thailand Malaysia Philippines
  7 weeks  3.672  3.443   2.747  3.609  2.955  3.611    4.502    3.558
  rep     12.000 10.000   6.000 16.000  6.000  8.000   18.000    6.000
  1 week   3.477  3.110   2.250  3.373  2.268  4.071    3.559    3.665
  rep     12.000 10.000   6.000 16.000  6.000  8.000   18.000    6.000
         country
inoc      Australia PNG    Solomon Is. Mauritius Sri Lanka Guam   China
  7 weeks  2.959     3.850  4.200       3.390     3.695     2.245  4.030
  rep      8.000     2.000  4.000       2.000     6.000     2.000  6.000
  1 week   2.304     3.450  3.197       2.855     2.792     2.440  3.342
```



```
  rep       8.000      2.000  4.000        2.000      6.000      2.000   6.000
          country
inoc       Puerto Rico Vanuatu Benin
  7 weeks  3.540        2.720   3.870
  rep      2.000        2.000   2.000
  1 week   3.150        2.805   2.815
  rep      2.000        2.000   2.000

 inoc:prov
          prov
inoc       1     2     3     4     5     6     7     8     10    11    12
  7 weeks 2.427 4.682 3.757 3.637 3.625 3.540 4.100 3.774 3.559 3.544 3.135
  rep     2.000 2.819 3.344 3.664 2.904 3.183 2.646 3.301 3.308 3.183 3.058
  1 week  2.819 3.344 3.664 2.904 3.183 2.646 3.301 3.308 3.183 3.058 3.225
  rep     2.427 4.682 3.757 3.637 3.625 3.540 4.100 3.774 3.559 3.544 3.135
          prov
inoc       13    14    15    16    17    18    19    20    21    22    23
  7 weeks 3.845 3.625 4.104 4.119 3.604 3.114 3.509 3.304 3.591 3.801 3.276
  rep     3.225 2.895 3.183 3.085 2.740 2.945 3.055 3.810 3.460 2.880 3.120
  1 week  2.895 3.183 3.085 2.740 2.945 3.055 3.810 3.460 2.880 3.120 2.885
  rep     3.845 3.625 4.104 4.119 3.604 3.114 3.509 3.304 3.591 3.801 3.276
          prov
inoc       24    25    26    27    28    29    30    31    32    33    34
  7 weeks 3.021 3.976 4.286 4.186 2.866 3.663 2.988 3.793 4.358 3.343 3.468
  rep     2.885 4.090 3.860 3.010 2.675 2.945 3.754 3.404 3.729 2.474 3.014
  1 week  4.090 3.860 3.010 2.675 2.945 3.754 3.404 3.729 2.474 3.014 2.449
  rep     3.021 3.976 4.286 4.186 2.866 3.663 2.988 3.793 4.358 3.343 3.468
          prov
inoc       35    36    37    38    39    40    41    42    45    46    47
  7 weeks 3.848 3.688 3.625 3.772 3.132 3.972 3.545 3.705 3.439 2.599 4.119
  rep     2.449 3.424 3.064 3.183 2.733 3.168 3.648 2.845 3.520 3.597 2.922
  1 week  3.424 3.064 3.183 2.733 3.168 3.648 2.845 3.520 3.597 2.922 3.347
  rep     3.848 3.688 3.625 3.772 3.132 3.972 3.545 3.705 3.439 2.599 4.119
          prov
inoc       48    50    51    52    53    54    55    56    57    58    59
  7 weeks 4.344 3.625 4.137 4.152 4.047 3.167 2.622 3.895 3.478 3.625 3.625
  rep     3.347 2.867 3.183 3.233 2.318 3.463 4.043 2.858 3.430 3.339 3.183
  1 week  2.867 3.183 3.233 2.318 3.463 4.043 2.858 3.430 3.339 3.183 3.183
  rep     4.344 3.625 4.137 4.152 4.047 3.167 2.622 3.895 3.478 3.625 3.625
          prov
inoc       60    61    62    63
  7 weeks 3.685 3.630 3.560 3.235
  rep     3.183 3.371 2.726 3.451
  1 week  3.371 2.726 3.451 3.601
  rep     3.685 3.630 3.560 3.235

 RESFit <-
     data.frame(
       fittedvalue     = fitted.aovlist(fm8.4)
     , residualvalue = proj(fm8.4)$Within[,"Residuals"]
     )

 ggplot(RESFit,aes(x=fittedvalue,y=residualvalue))+
 geom_point(size=2)+
 labs(x="Residuals vs Fitted Values", y="")+
 theme_bw()
```




```
# Pg. 153
fm8.6 <-
 aov(
   formula = terms(dbh ~ inoc + repl + col + repl:row + repl:col +
                         prov + inoc:prov, keep.order = TRUE)
 , data    = DataExam8.1
 )

summary(fm8.6)
            Df Sum Sq Mean Sq F value                Pr(>F)
inoc         1  11.54  11.542  48.054        0.00000000327 ***
repl         2   2.01   1.007   4.193             0.019746 *
col          9  65.24   7.249  30.182 < 0.0000000000000002 ***
repl:row    20  16.59   0.830   3.454             0.000105 ***
repl:col    27  16.41   0.608   2.530             0.001443 **
prov        58  53.89   0.929   3.869        0.00000026687 ***
inoc:prov   58   8.47   0.146   0.608             0.970544
Residuals   60  14.41   0.240
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

8.2 Example 8.1 (continued) (Pg. 155)

Example 8.1 (continued) (Pg. 155)

In the early 1990s Khongsak Pinyopusarerk of CSIRO Forestry and Forest Products initiated a far-reaching study of Casuarina equisetifolia. This is a nitrogen-fixing tree of considerable social, economic and environmental importance in tropical/ subtropical littoral zones of Asia, the Pacific and Africa. Provenance collections were obtained from 18 countries and, with this material, more than 40 trials were laid out in 20 countries. The number of seedlots included in each trial varied, depending on the suitability and size

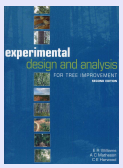
of the planting sites for the available material. One of the trials, in Weipa, northern Queensland, contained all the available seedlots and is the example used here. Further details are given by Pinyopusarerk & Williams (2000).

```
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam8.1)

# Pg. 155
fm8.8 <-
    lmerTest::lmer(
            formula     = dbh ~ 1 + repl + col + prov + (1|repl:row) + (1|repl:col)
          , data        = DataExam8.1
          , REML        = TRUE
          )

# Pg. 157
#\dontrun{
varcomp(fm8.8)
                      vcov     SE  2.5 % 97.5 %
repl:col.(Intercept) 0.0459 0.0262 0.0000 0.0565
repl:row.(Intercept) 0.0640 0.0294 0.0210 0.1161
residual             0.1951 0.0253 0.1126 0.1782

#}

anova(fm8.8)
Type III Analysis of Variance Table with Satterthwaite's method
     Sum Sq Mean Sq NumDF   DenDF F value              Pr(>F)
repl  2.581 0.86023     3  21.257  4.4082             0.01469 *
col  24.874 2.76378     9  23.705 14.1627 0.00000015112494811 ***
prov 55.433 0.95574    58 136.623  4.8976 0.00000000000001306 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

anova(fm8.8, ddf = "Kenward-Roger")
Type III Analysis of Variance Table with Kenward-Roger's method
     Sum Sq Mean Sq NumDF   DenDF F value             Pr(>F)
repl  2.580 0.86016     3  22.622  4.4078             0.0139 *
col  24.824 2.75827     9  22.947 14.1337 0.0000002097637857 ***
prov 54.795 0.94473    58 133.852  4.8396 0.0000000000000283 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 predictmeans(model = fm8.8, modelterm = "repl")
```

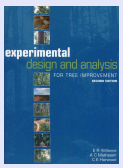
```
$`Predicted Means`
repl
     1      2      3      4
3.7543 3.1269 3.2128 3.5023

$`Standard Error of Means`
All means have the same SE
                   0.13629

$`Standard Error of Differences`
  Max.SED   Min.SED  Aveg.SED
0.1927495 0.1927423 0.1927472

$LSD
 Max.LSD  Min.LSD Aveg.LSD
 0.39910  0.39909  0.39910
attr(,"Significant level")
[1] 0.05
attr(,"Degree of freedom")
[1] 22.62

$mean_table
  repl   Mean      SE       Df LL(95%) UL(95%)
1    1 3.7543 0.13629 22.62064  3.4721  4.0365
2    2 3.1269 0.13629 22.62064  2.8447  3.4091
3    3 3.2128 0.13629 22.62064  2.9306  3.4950
4    4 3.5023 0.13629 22.62064  3.2201  3.7845


 predictmeans(model = fm8.8, modelterm = "col")
```

```
$`Predicted Means`
col
     1      2      3      4      5      6      7      8      9     10
3.5053 3.4996 3.8509 3.8280 3.5947 3.7829 3.3059 3.5158 3.1776 1.9301

$`Standard Error of Means`
col
      1       2       3       4       5       6       7       8       9      10
0.15496 0.15649 0.15493 0.15494 0.15490 0.15721 0.15479 0.15699 0.15485 0.15648

$`Standard Error of Differences`
  Max.SED   Min.SED  Aveg.SED
0.2132295 0.2061223 0.2086161

$LSD
 Max.LSD  Min.LSD Aveg.LSD
 0.43749  0.42291  0.42802
attr(,"Significant level")
[1] 0.05
attr(,"Degree of freedom")
[1] 27.03

$mean_table
   col   Mean      SE       Df LL(95%) UL(95%)
```

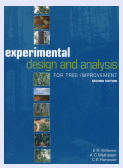


```
1     1 3.5053 0.15496 27.03111  3.1873  3.8232
2     2 3.4996 0.15649 27.03111  3.1786  3.8207
3     3 3.8509 0.15493 27.03111  3.5330  4.1687
4     4 3.8280 0.15494 27.03111  3.5101  4.1459
5     5 3.5947 0.15490 27.03111  3.2769  3.9125
6     6 3.7829 0.15721 27.03111  3.4604  4.1055
7     7 3.3059 0.15479 27.03111  2.9883  3.6234
8     8 3.5158 0.15699 27.03111  3.1937  3.8379
9     9 3.1776 0.15485 27.03111  2.8599  3.4953
10   10 1.9301 0.15648 27.03111  1.6091  2.2512


 predictmeans(model = fm8.8, modelterm = "prov")
```

```
$`Predicted Means`
prov
     1      2      3      4      5      6      7      8     10     11     12
2.4222 3.1425 2.8646 2.2599 3.7189 3.5445 3.9474 2.7469 2.6459 2.0497 2.7497
    13     14     15     16     17     18     19     20     21     22     23
2.9833 2.6534 3.4459 3.9209 3.4294 3.3624 3.6540 3.3685 3.2210 3.3315 3.2453
    24     25     26     27     28     29     30     31     32     33     34
3.6933 3.8460 3.5233 3.5271 3.1001 3.8997 3.5447 4.1294 4.2755 3.7429 3.8890
    35     36     37     38     39     40     41     42     45     46     47
4.1179 3.8526 3.4538 3.0379 3.5306 3.7412 3.7422 3.8765 4.2126 3.4660 4.4688
    48     50     51     52     53     54     55     56     57     58     59
3.7503 2.5571 3.4571 3.3227 3.5926 3.5337 2.7157 2.7702 3.9805 2.9968 3.3978
    60     61     62     63
3.0115 3.2238 3.2814 3.5751

$`Standard Error of Means`
prov
      1       2       3       4       5       6       7       8      10      11
0.25362 0.25446 0.25278 0.25443 0.25423 0.25588 0.25420 0.25338 0.25417 0.25259
     12      13      14      15      16      17      18      19      20      21
0.25306 0.25436 0.25259 0.25232 0.25502 0.25319 0.25340 0.25309 0.25294 0.25204
     22      23      24      25      26      27      28      29      30      31
0.25350 0.25355 0.25358 0.25316 0.25289 0.25294 0.25291 0.25341 0.25254 0.25330
     32      33      34      35      36      37      38      39      40      41
0.25377 0.25387 0.25510 0.25347 0.25351 0.25352 0.25416 0.25393 0.25450 0.25416
     42      45      46      47      48      50      51      52      53      54
0.25303 0.25257 0.25323 0.25316 0.25356 0.25354 0.25372 0.25361 0.25237 0.25321
     55      56      57      58      59      60      61      62      63
0.25374 0.25639 0.25173 0.25319 0.25250 0.25303 0.25378 0.25266 0.25500

$`Standard Error of Differences`
  Max.SED   Min.SED  Aveg.SED
0.3591712 0.3326719 0.3482665

$LSD
 Max.LSD  Min.LSD Aveg.LSD
 0.70929  0.65696  0.68775
attr(,"Significant level")
[1] 0.05
attr(,"Degree of freedom")
[1] 161.14

$mean_table
   prov   Mean       SE        Df LL(95%) UL(95%)
1     1 2.4222 0.25362 161.1392  1.9214  2.9231
```



2	2	3.1425	0.25446	161.1392	2.6400	3.6450
3	3	2.8646	0.25278	161.1392	2.3654	3.3638
4	4	2.2599	0.25443	161.1392	1.7574	2.7623
5	5	3.7189	0.25423	161.1392	3.2168	4.2209
6	6	3.5445	0.25588	161.1392	3.0392	4.0498
7	7	3.9474	0.25420	161.1392	3.4454	4.4494
8	8	2.7469	0.25338	161.1392	2.2466	3.2473
9	10	2.6459	0.25417	161.1392	2.1440	3.1478
10	11	2.0497	0.25259	161.1392	1.5509	2.5485
11	12	2.7497	0.25306	161.1392	2.2499	3.2494
12	13	2.9833	0.25436	161.1392	2.4809	3.4856
13	14	2.6534	0.25259	161.1392	2.1546	3.1522
14	15	3.4459	0.25232	161.1392	2.9476	3.9442
15	16	3.9209	0.25502	161.1392	3.4173	4.4245
16	17	3.4294	0.25319	161.1392	2.9294	3.9294
17	18	3.3624	0.25340	161.1392	2.8620	3.8628
18	19	3.6540	0.25309	161.1392	3.1542	4.1538
19	20	3.3685	0.25294	161.1392	2.8690	3.8680
20	21	3.2210	0.25204	161.1392	2.7233	3.7188
21	22	3.3315	0.25350	161.1392	2.8309	3.8321
22	23	3.2453	0.25355	161.1392	2.7446	3.7461
23	24	3.6933	0.25358	161.1392	3.1926	4.1941
24	25	3.8460	0.25316	161.1392	3.3460	4.3459
25	26	3.5233	0.25289	161.1392	3.0239	4.0227
26	27	3.5271	0.25294	161.1392	3.0276	4.0266
27	28	3.1001	0.25291	161.1392	2.6007	3.5996
28	29	3.8997	0.25341	161.1392	3.3993	4.4001
29	30	3.5447	0.25254	161.1392	3.0460	4.0434
30	31	4.1294	0.25330	161.1392	3.6292	4.6297
31	32	4.2755	0.25377	161.1392	3.7744	4.7767
32	33	3.7429	0.25387	161.1392	3.2415	4.2442
33	34	3.8890	0.25510	161.1392	3.3853	4.3928
34	35	4.1179	0.25347	161.1392	3.6174	4.6185
35	36	3.8526	0.25351	161.1392	3.3520	4.3533
36	37	3.4538	0.25352	161.1392	2.9531	3.9544
37	38	3.0379	0.25416	161.1392	2.5360	3.5398
38	39	3.5306	0.25393	161.1392	3.0291	4.0320
39	40	3.7412	0.25450	161.1392	3.2386	4.2437
40	41	3.7422	0.25416	161.1392	3.2403	4.2441
41	42	3.8765	0.25303	161.1392	3.3768	4.3762
42	45	4.2126	0.25257	161.1392	3.7138	4.7114
43	46	3.4660	0.25323	161.1392	2.9660	3.9661
44	47	4.4688	0.25316	161.1392	3.9688	4.9687
45	48	3.7503	0.25356	161.1392	3.2496	4.2510
46	50	2.5571	0.25354	161.1392	2.0564	3.0578
47	51	3.4571	0.25372	161.1392	2.9561	3.9582
48	52	3.3227	0.25361	161.1392	2.8219	3.8236
49	53	3.5926	0.25237	161.1392	3.0942	4.0910
50	54	3.5337	0.25321	161.1392	3.0337	4.0338
51	55	2.7157	0.25374	161.1392	2.2146	3.2167
52	56	2.7702	0.25639	161.1392	2.2639	3.2765
53	57	3.9805	0.25173	161.1392	3.4834	4.4776
54	58	2.9968	0.25319	161.1392	2.4968	3.4968
55	59	3.3978	0.25250	161.1392	2.8991	3.8964
56	60	3.0115	0.25303	161.1392	2.5118	3.5112
57	61	3.2238	0.25378	161.1392	2.7227	3.7250
58	62	3.2814	0.25266	161.1392	2.7825	3.7804
59	63	3.5751	0.25500	161.1392	3.0716	4.0787



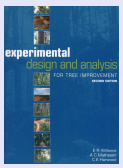
```
  # Pg. 161
   RCB1  <- aov(dbh ~ prov + repl, data = DataExam8.1)

   RCB   <- emmeans(RCB1,  specs = "prov") %>% as_tibble()

   Mixed <- emmeans(fm8.8, specs = "prov") %>% as_tibble()

   table8.9 <- left_join(RCB, Mixed, by = "prov", suffix = c(".RCBD", ".Mixed"))

   print(table8.9)
# A tibble: 59 x 11
   prov  emmean.RCBD SE.RCBD df.RCBD lower.CL.RCBD upper.CL.RCBD emmean.Mixed
   <fct>       <dbl>   <dbl>   <dbl>         <dbl>         <dbl>        <dbl>
 1 1            1.85   0.382     174          1.10          2.60         2.42
 2 2            3.24   0.382     174          2.49          3.99         3.14
 3 3            2.94   0.382     174          2.18          3.69         2.86
 4 4            2.50   0.382     174          1.74          3.25         2.26
 5 5            3.34   0.382     174          2.59          4.10         3.72
 6 6            3.37   0.382     174          2.62          4.13         3.54
 7 7            3.98   0.382     174          3.23          4.74         3.95
 8 8            2.63   0.382     174          1.88          3.39         2.75
 9 10           2.47   0.382     174          1.71          3.22         2.65
10 11           2.40   0.382     174          1.64          3.15         2.05
# i 49 more rows
# i 4 more variables: SE.Mixed <dbl>, df.Mixed <dbl>, lower.CL.Mixed <dbl>,
#   upper.CL.Mixed <dbl>
```

8.3 Example 8.1 (continued) (Pg. 167)

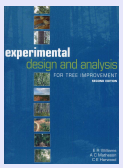
Example 8.1 (continued) (Pg. 167)

The program in Table 8.7 creates a file of estimated provenance means, namely Tab8_9.mns. These estimated means can then be analysed according to the provenances-within-countries information given in Table 8.1. The GenStat program to analyse this nested seedlot structure is given in Table 8.10. The resulting analysis of variance table is presented in Table 8.11; the residual mean square (0.1951) from the analysis in Table 8.8 has been manually added to this table. Remember that we carried out a similar operation in Table 5.7 when adopting a multi-stage approach to analysis across sites.

```
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)
```



```
library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam8.1)

# Pg. 167
fm8.11 <- aov(formula = dbh ~ country + country:prov, data = DataExam8.1)

  b <- anova(fm8.11)

  Res                      <- length(b[["Sum Sq"]])

  df                       <- 119

  MSS                      <- 0.1951

  b[["Df"]][Res]           <- df

  b[["Sum Sq"]][Res]       <- MSS*df

  b[["Mean Sq"]][Res]      <- b[["Sum Sq"]][Res]/b[["Df"]][Res]

  b[["F value"]][1:Res-1] <- b[["Mean Sq"]][1:Res-1]/b[["Mean Sq"]][Res]

  b[["Pr(>F)"]][Res-1]      <- df(b[["F value"]][Res-1],b[["Df"]][Res-1],b[["Df"]][Res])

  b
Analysis of Variance Table

Response: dbh
              Df Sum Sq Mean Sq F value        Pr(>F)
country       17 54.619  3.2129  16.468 0.00000001235 ***
country:prov  41 18.606  0.4538   2.326      0.001502 **
Residuals    119 23.217  0.1951
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

  emmeans(fm8.11, specs = "country")
 country     emmean    SE  df lower.CL upper.CL
 India         3.57 0.165 177     3.25     3.90
 Vietnam       3.28 0.181 177     2.92     3.63
 Egypt         2.50 0.233 177     2.04     2.96
 Kenya         3.49 0.143 177     3.21     3.77
 Fiji          2.61 0.233 177     2.15     3.07
 Thailand      3.84 0.202 177     3.44     4.24
 Malaysia      4.03 0.135 177     3.77     4.30
 Philippines   3.61 0.233 177     3.15     4.07
 Australia     2.63 0.202 177     2.23     3.03
 PNG           3.65 0.404 177     2.85     4.45
 Solomon Is.   3.70 0.285 177     3.14     4.26
 Mauritius     3.12 0.404 177     2.33     3.92
 Sri Lanka     3.24 0.233 177     2.78     3.70
 Guam          2.34 0.404 177     1.55     3.14
 China         3.69 0.233 177     3.23     4.15
 Puerto Rico   3.35 0.404 177     2.55     4.14
 Vanuatu       2.76 0.404 177     1.97     3.56
```



```
 Benin          3.34 0.404 177     2.55     4.14

Results are averaged over the levels of: prov
Confidence level used: 0.95
```

8.4 Example 8.2 (Pg.)

Example 8.2 (Pg.)

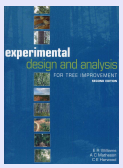
Statement

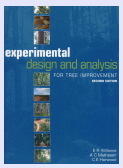
```
library(car)

library(dae)

library(dplyr)

library(emmeans)

library(ggplot2)

library(lmerTest)

library(magrittr)

library(predictmeans)

library(supernova)

data(DataExam8.2)

# Pg.
fm8.2  <-
  lmer(
    formula = dbhmean ~ Repl + Column + Contcompf + Contcompf:Standard +
              (1|Repl:Row ) + (1|Repl:Column ) + (1|Contcompv:Clone)
  , data    = DataExam8.2
    )

#\dontrun{
varcomp(fm8.2)
                              vcov     SE  2.5 % 97.5 %
Contcompv:Clone.(Intercept) 0.4950 0.1126 0.3057 0.7422
Repl:Row.(Intercept)        0.0802 0.0351 0.0173 0.1458
Repl:Column.(Intercept)     0.0529 0.0326 0.0000 0.0783
residual                    0.3992 0.0435 0.3245 0.5024

#}
anova(fm8.2)
Type III Analysis of Variance Table with Satterthwaite's method
                    Sum Sq Mean Sq NumDF   DenDF F value          Pr(>F)
Repl                3.2720  0.8180     4  26.467  2.0489       0.1162606
Column              3.1018  0.6204     5  19.545  1.5539       0.2194719
Contcompf           5.3203  5.3203     1  54.905 13.3265       0.0005845 ***
```

```
Contcompf:Standard 20.6587  6.8862     3 207.152 17.2488 0.0000000004896 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 Anova(fm8.2, type = "II", test.statistic = "Chisq")
Analysis of Deviance Table (Type II Wald chisquare tests)

Response: dbhmean
                     Chisq Df Pr(>Chisq)
Repl                8.1957  4    0.08467 .
Column              7.7694  5    0.16941
Contcompf           4.6841  1    0.03044 *
Contcompf:Standard 51.7463  3  3.392e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 predictmeans(model = fm8.2, modelterm = "Repl")
```

```
$`Predicted Means`
Repl
     1      2      3      4      5
7.8926 8.2070 8.3429 8.4604 8.5464

$`Standard Error of Means`
Repl
      1       2       3       4       5
0.33123 0.33126 0.32992 0.32992 0.32992

$`Standard Error of Differences`
  Max.SED   Min.SED  Aveg.SED
0.2239675 0.2167320 0.2196681

$LSD
 Max.LSD  Min.LSD Aveg.LSD
 0.44792  0.43345  0.43932
attr(,"Significant level")
[1] 0.05
attr(,"Degree of freedom")
[1] 60.56

$mean_table
  Repl   Mean      SE       Df LL(95%) UL(95%)
1    1 7.8926 0.33123 60.55892  7.2302  8.5551
2    2 8.2070 0.33126 60.55892  7.5445  8.8695
3    3 8.3429 0.32992 60.55892  7.6831  9.0027
4    4 8.4604 0.32992 60.55892  7.8006  9.1202
5    5 8.5464 0.32992 60.55892  7.8866  9.2062


 predictmeans(model = fm8.2, modelterm = "Column")
```

```
$`Predicted Means`
Column
     1      2      3      4      5      6
8.2214 8.4708 8.3779 7.9721 7.8166 8.7141

$`Standard Error of Means`
Column
      1       2       3       4       5       6
```



```
0.31662 0.39168 0.39315 0.26648 0.26646 0.31653

$`Standard Error of Differences`
  Max.SED   Min.SED  Aveg.SED
0.2714760 0.2102583 0.2373610

$LSD
 Max.LSD  Min.LSD Aveg.LSD
 0.54413  0.42143  0.47575
attr(,"Significant level")
[1] 0.05
attr(,"Degree of freedom")
[1] 54.65

$mean_table
  Column   Mean      SE       Df LL(95%) UL(95%)
1      1 8.2214 0.31662 54.64679  7.5868  8.8561
2      2 8.4708 0.39168 54.64679  7.6857  9.2558
3      3 8.3779 0.39315 54.64679  7.5900  9.1659
4      4 7.9721 0.26648 54.64679  7.4380  8.5063
5      5 7.8166 0.26646 54.64679  7.2825  8.3507
6      6 8.7141 0.31653 54.64679  8.0797  9.3486


 library(emmeans)

 emmeans(object = fm8.2, specs = ~Contcompf|Standard)
Contcompf = 1, Standard = 0:
 emmean    SE   df lower.CL upper.CL
   8.91 0.117 65.9     8.67     9.14

Contcompf = 0, Standard = UG323:
 emmean    SE   df lower.CL upper.CL
   8.97 0.770 55.6     7.43    10.51

Contcompf = 0, Standard = U6:
 emmean    SE   df lower.CL upper.CL
   6.55 0.770 55.5     5.01     8.10

Contcompf = 0, Standard = PN14:
 emmean    SE   df lower.CL upper.CL
   7.70 0.771 55.8     6.16     9.25

Contcompf = 0, Standard = SSOseed:
 emmean    SE   df lower.CL upper.CL
   6.08 0.770 55.5     4.54     7.63

Results are averaged over the levels of: Repl, Column
Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95
```

References